

Module 09: ER-to-Relational Mapping & Table Reduction

1. The 7 Conversion Rules

ER-to-Relational Schema Conversion Rules

The 7 Fundamental Mapping Algorithms and Relational Table Minimization

Rule 1: Strong Entity Set → 1 Table
All simple attributes become columns. Composite attributes flattened.

STUDENT(RollNo, Name, Street, City, State, DOB)

Rule 2: Weak Entity Set → 1 Table (With Owner PK)
Weak attributes + Owner PK as Foreign Key. PK is Composite.

DEPENDENT(Emp_ID (FK), Dep_Name, Relation, Age)

Rule 3: Binary 1:1 Relationship
• Foreign Key approach: Add PK of either side as FK (prefer Total side).
• Merged Table approach: If BOTH sides Total, combine into 1 table!

CITIZEN(Aadhar_No, Name) | PASSPORT(Pass_No, Issue, Aadhar_No(FK))

Rule 4: Binary 1:N Relationship → Foreign Key on 'N' Side
Do NOT create separate table! Place PK of '1' side into 'N' side table.

DEPT(Dept_ID, DName) [1 Side]
EMPLOYEE(Emp_ID, EName, Dept_ID (FK)) [N Side]

Rule 5: Binary M:N Relationship → Dedicated Join Table
Requires separate relationship table! Primary Key is composite.

STUDENT(RollNo, Name) | COURSE(Course_ID, Title)
ENROLLED(RollNo(FK), Course_ID(FK), Grade, Enroll_Date)

Rule 6: Multivalued Attribute → Separate Table
Stores Entity PK + Multivalued Attribute. PK is composite pair.

STUDENT(RollNo, Name)
STUDENT_PHONE(RollNo(FK), Phone_Number)

Table Minimization Principles (Exam Formula):

- 1:1 with Both Total → 1 Table (Merge both entities + relation)
- 1:1 with One Partial → 2 Tables (Foreign key on Total side)
- 1:N Relationship → 2 Tables (Merge relation into 'N' side)
- M:N Relationship → 3 Tables (Entity A, Entity B, and Join Table)
- Weak Entity → 2 Tables (Merged identifying relationship)

Golden Rule: Never create separate tables for 1:N relationships; always merge with the 'N' side entity table!

2. Quick Minimization Formula (AKTU Numerical)

Relationship Type	Participation	Minimum Relational Tables
1 : 1	Both Total Participation	1 Table (Fully Merged)
1 : 1	One Total, One Partial	2 Tables (FK on Total side)
1 : N	Any Participation	2 Tables (Merge relation into 'N' side)
M : N	Any Participation	3 Tables (Table A, Table B + Join Table)
Weak Entity	Total Participation (Standard)	2 Tables (Owner Table + Weak Table)
Multivalued Attribute	1 Multivalued column	2 Tables (Entity Table + Attribute Table)